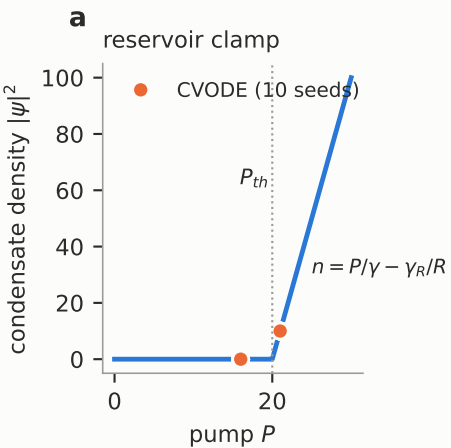
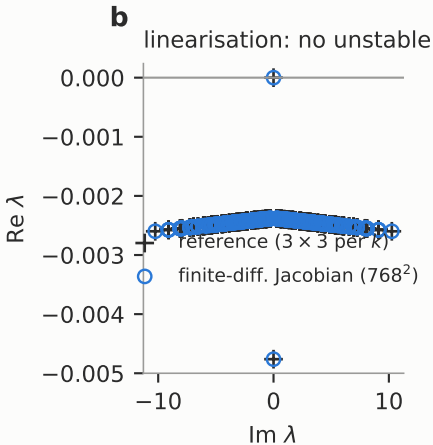
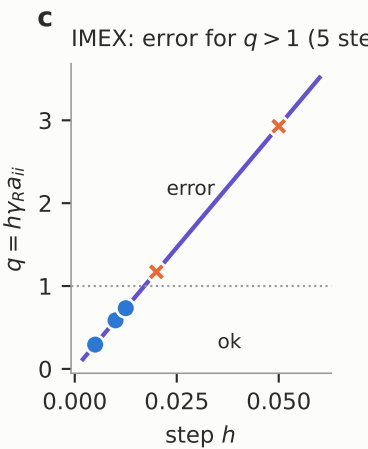

a
reservoir clamp
CVODE (10 seeds)
P_{th}
$n = P/\gamma - \gamma_R/R$
condensate density $|\psi|^2$
pump P
0
20
40
60
80
100


b
linearisation: no unstable mode
reference (3 × 3 per k)
finite-diff. Jacobian (768^2)
Re λ
Im λ
0.000
−0.001
−0.002
−0.003
−0.004
−0.005
−10
0
10


c
IMEX: error for $q > 1$ (5 ste
error
ok
$q = h\gamma_R a_{ii}$
step h
0
1
2
3
0.000
0.025
0.050